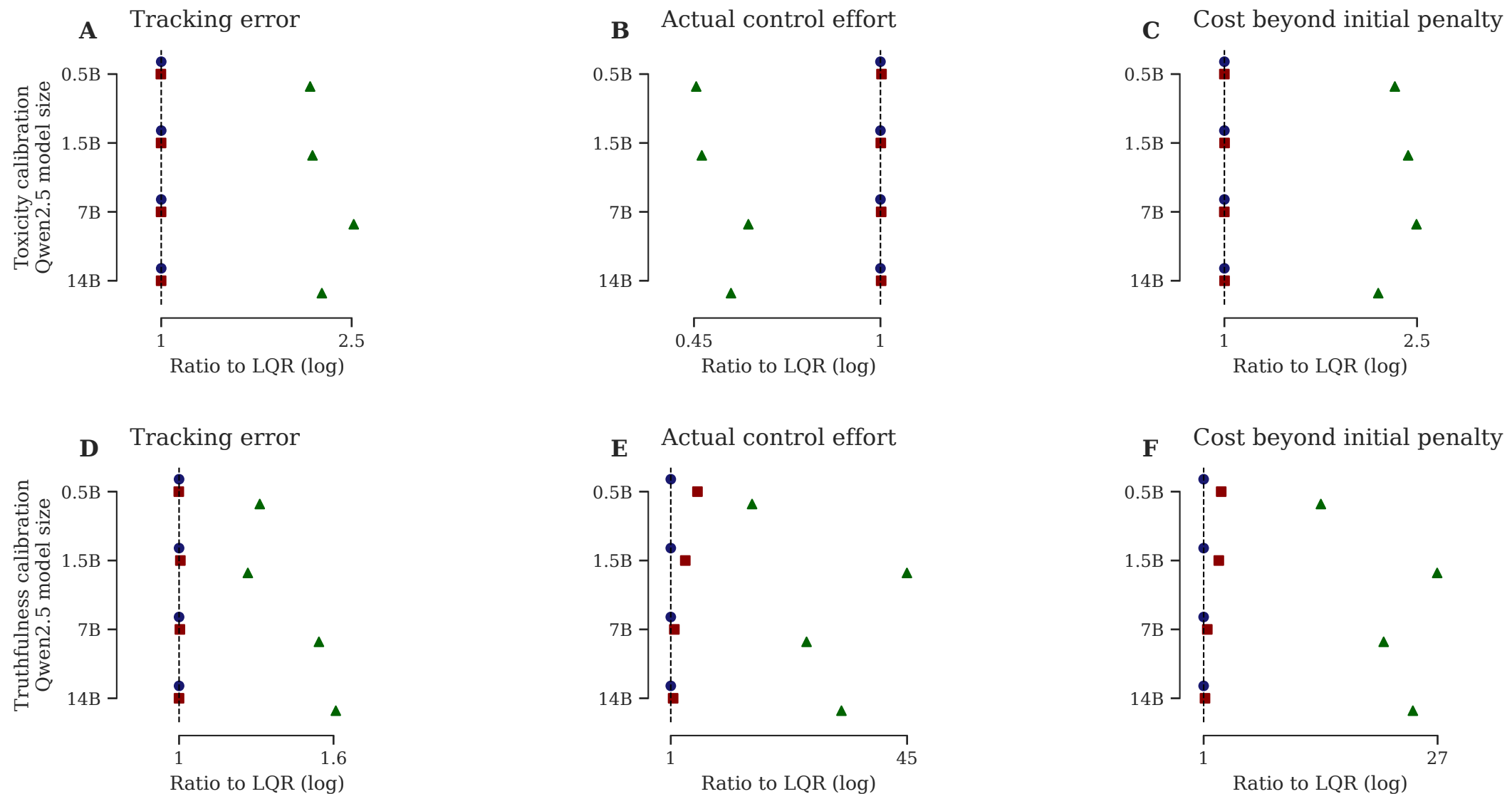


Controller comparison across Qwen models

Offline calibration replay · 50 matched paths per model and behavior · values relative to LQR

— LQR - - - H_∞ ···· PID (saved P-only tuning)



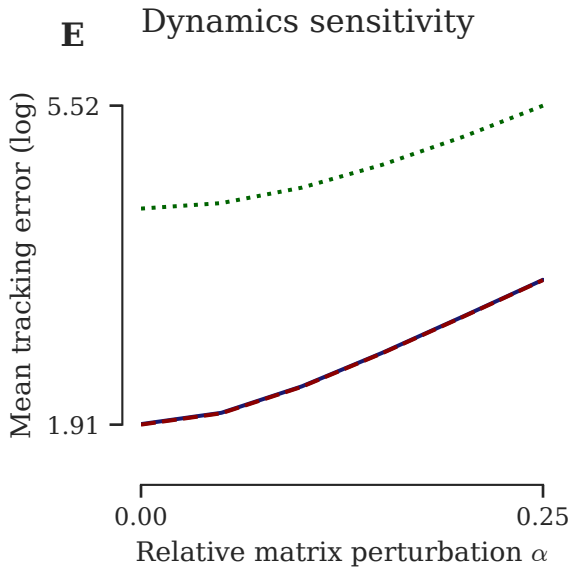
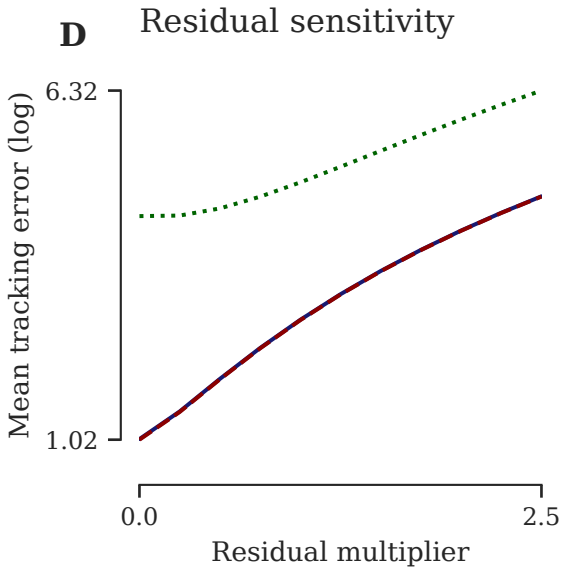
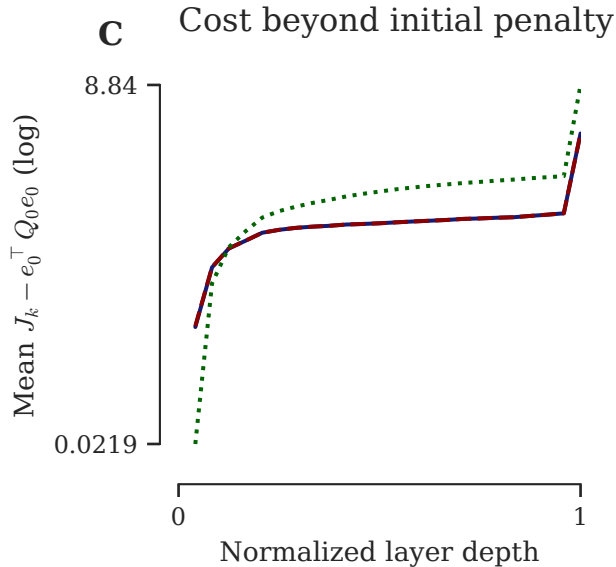
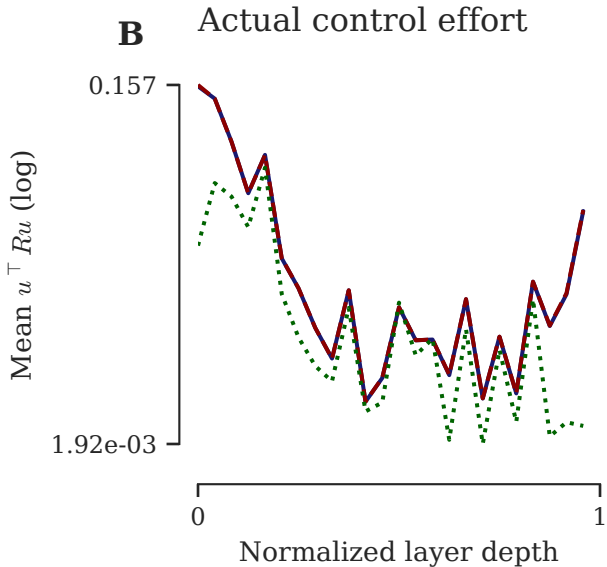
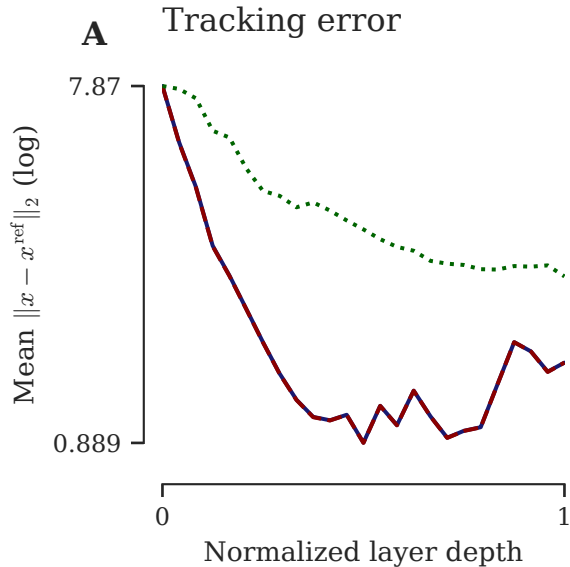
Below 1: lower than LQR. Same saved gains and matched inputs within each run.

Cost ratios exclude the common fixed initial-state penalty; full total costs are retained in the summary CSV.

Qwen2.5 0.5B · Toxicity calibration

Offline calibration replay · 50 matched paths · 24 layers · saved gains

— LQR - - - H_∞ ···· PID (saved P-only tuning)



F Replay summary

	LQR	H_∞	PID
Mean error	1.91	1.91	3.92
Maximum error	9.95	9.95	10.1
Effort J_u	0.662	0.665	0.301
Total cost	4.19	4.19	9.1
Beyond initial	3.93	3.93	8.84
Flagged paths	0/50	0/50	0/50

Beyond initial: $H_\infty/LQR = 1.000\times$
Lower cost is better.

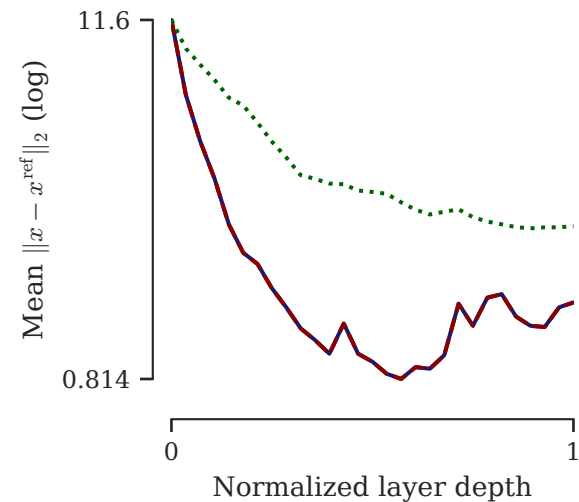
C subtracts the same fixed initial-state penalty for all controllers; full totals remain in F and the CSVs.
Costs include reference feedforward. Offline calibration replay; not recorded three-controller LM performance.

Qwen2.5 1.5B · Toxicity calibration

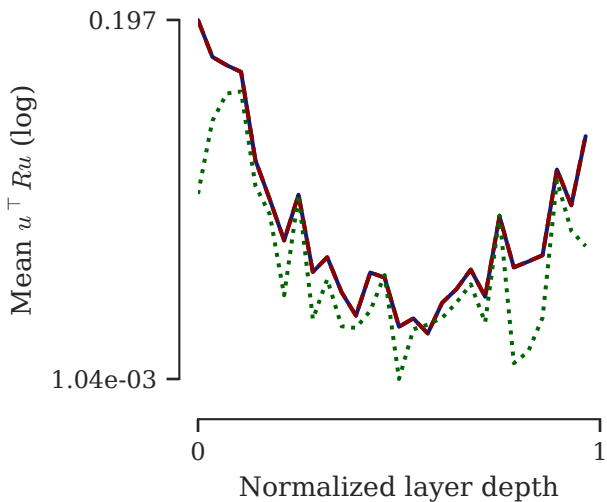
Offline calibration replay · 50 matched paths · 28 layers · saved gains

— LQR - - - H_∞ ···· PID (saved P-only tuning)

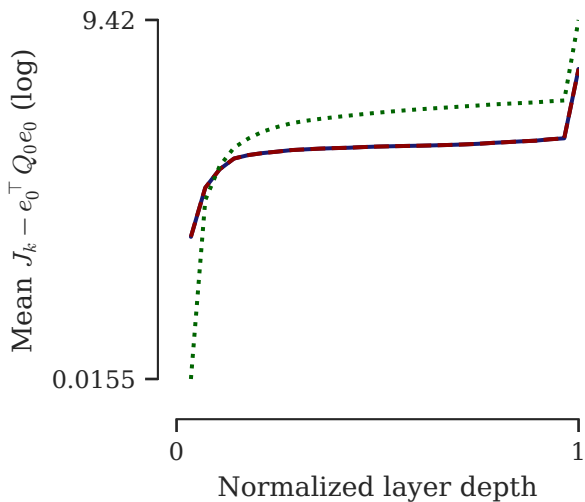
A Tracking error



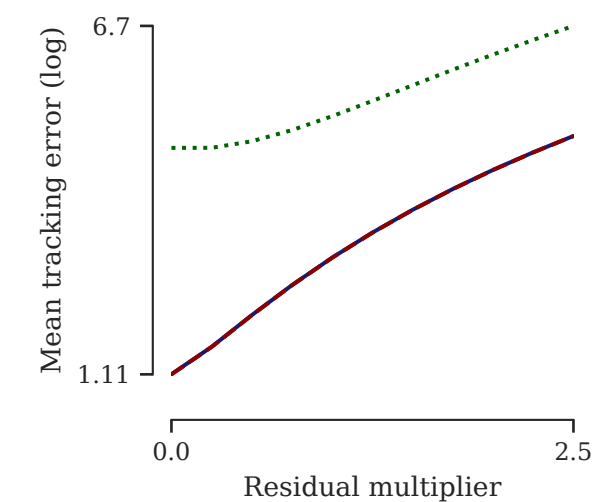
B Actual control effort



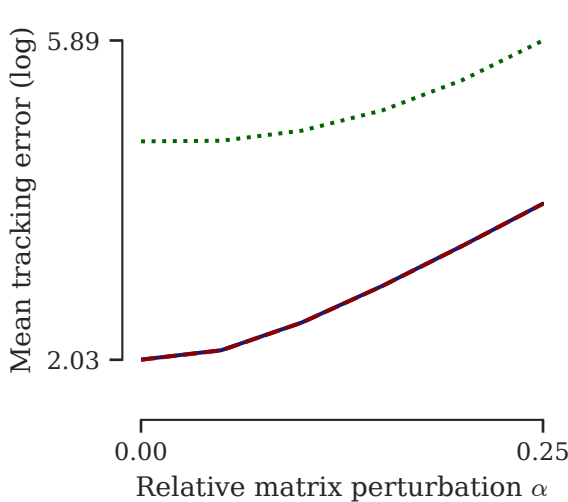
C Cost beyond initial penalty



D Residual sensitivity



E Dynamics sensitivity



F Replay summary

	LQR	H_∞	PID
Mean error	2.03	2.03	4.21
Maximum error	13.3	13.3	13.3
Effort J_u	0.716	0.717	0.333
Total cost	4.41	4.41	9.9
Beyond initial	3.93	3.93	9.42
Flagged paths	0/50	0/50	0/50

Beyond initial: $H_\infty/LQR = 1.001\times$
Lower cost is better.

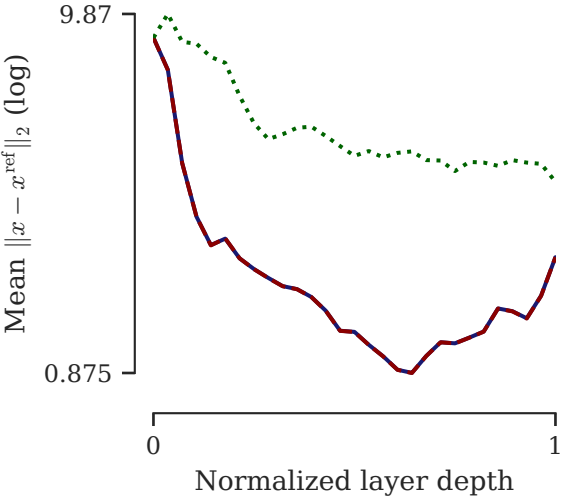
C subtracts the same fixed initial-state penalty for all controllers; full totals remain in F and the CSVs.
Costs include reference feedforward. Offline calibration replay; not recorded three-controller LM performance.

Qwen2.5 7B · Toxicity calibration

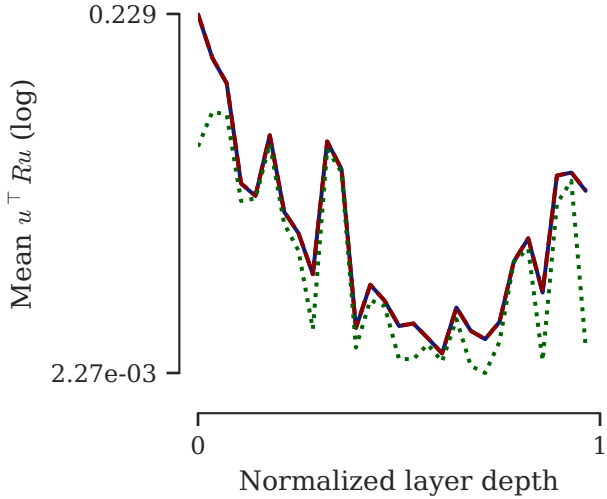
Offline calibration replay · 50 matched paths · 28 layers · saved gains

— LQR - - - H_∞ ···· PID (saved P-only tuning)

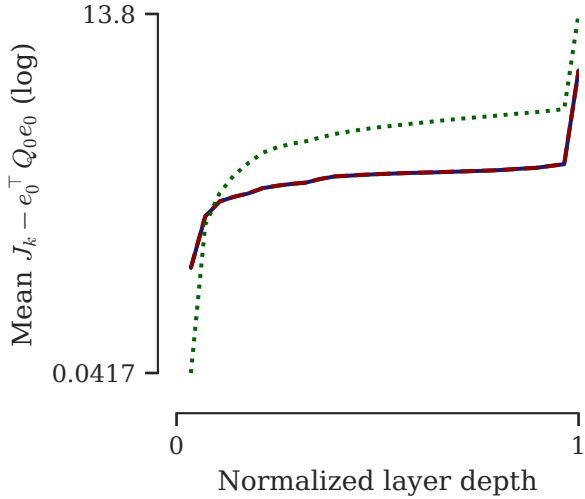
A Tracking error



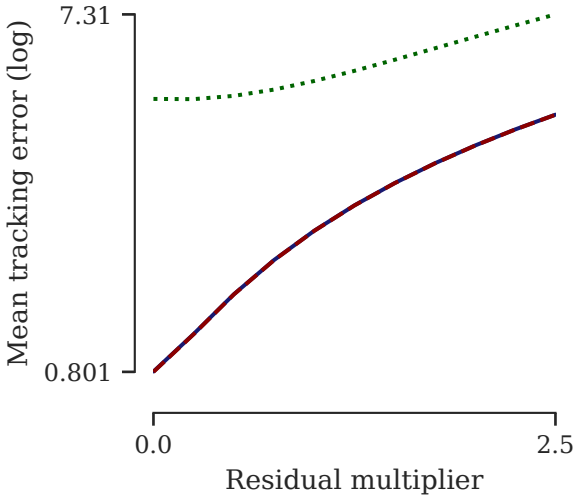
B Actual control effort



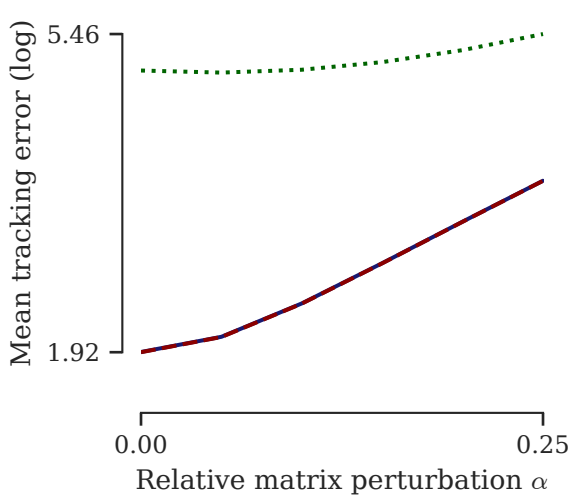
C Cost beyond initial penalty



D Residual sensitivity



E Dynamics sensitivity



F Replay summary

	LQR	H_∞	PID
Mean error	1.92	1.92	4.84
Maximum error	11	11	12
Effort J_u	0.823	0.825	0.467
Total cost	5.81	5.8	14.1
Beyond initial	5.55	5.55	13.8
Flagged paths	0/50	0/50	0/50

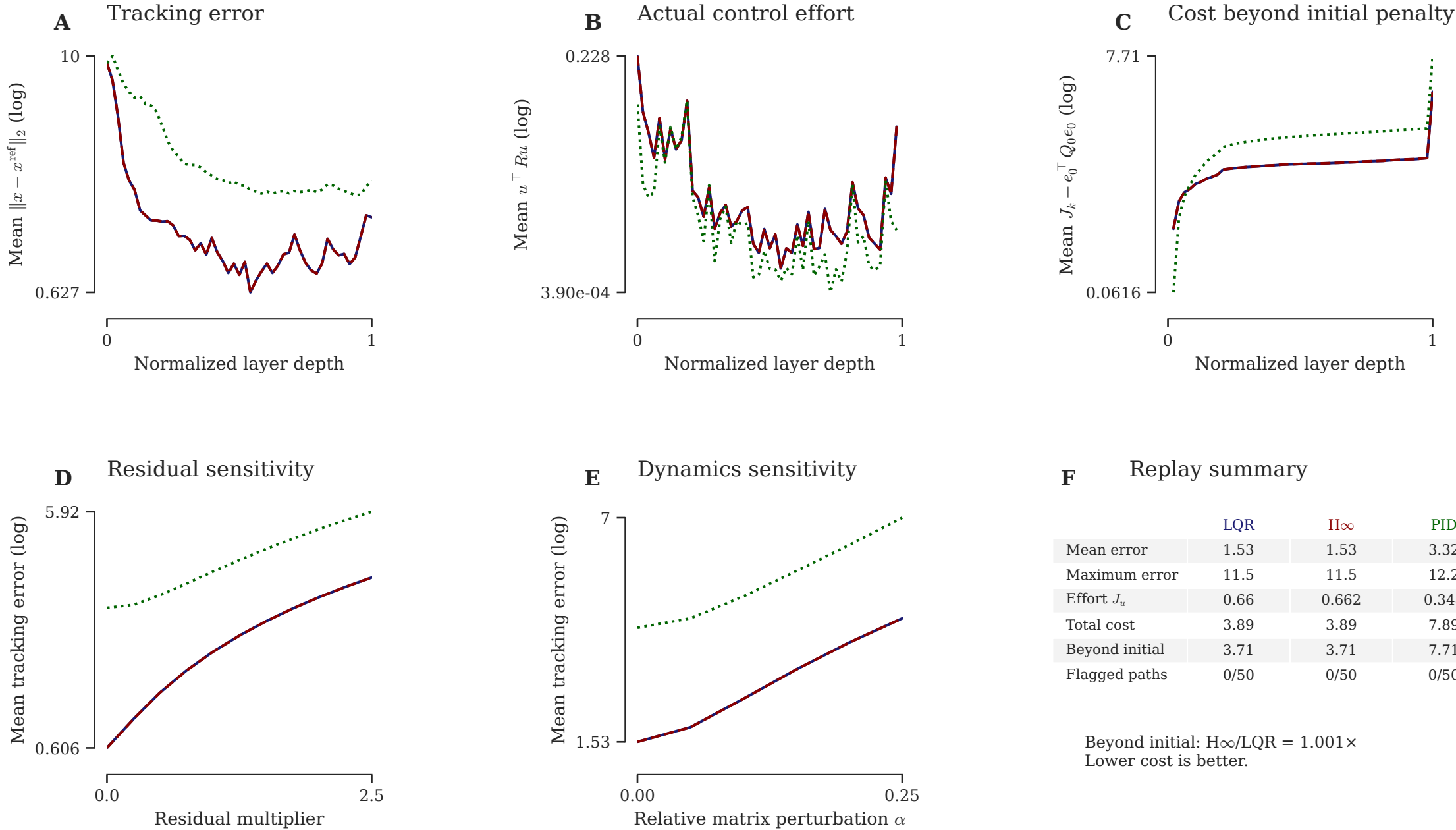
Beyond initial: $H_\infty/LQR = 1.000\times$
Lower cost is better.

C subtracts the same fixed initial-state penalty for all controllers; full totals remain in F and the CSVs.
Costs include reference feedforward. Offline calibration replay; not recorded three-controller LM performance.

Qwen2.5 14B · Toxicity calibration

Offline calibration replay · 50 matched paths · 48 layers · saved gains

— LQR - - - H_∞ ···· PID (saved P-only tuning)

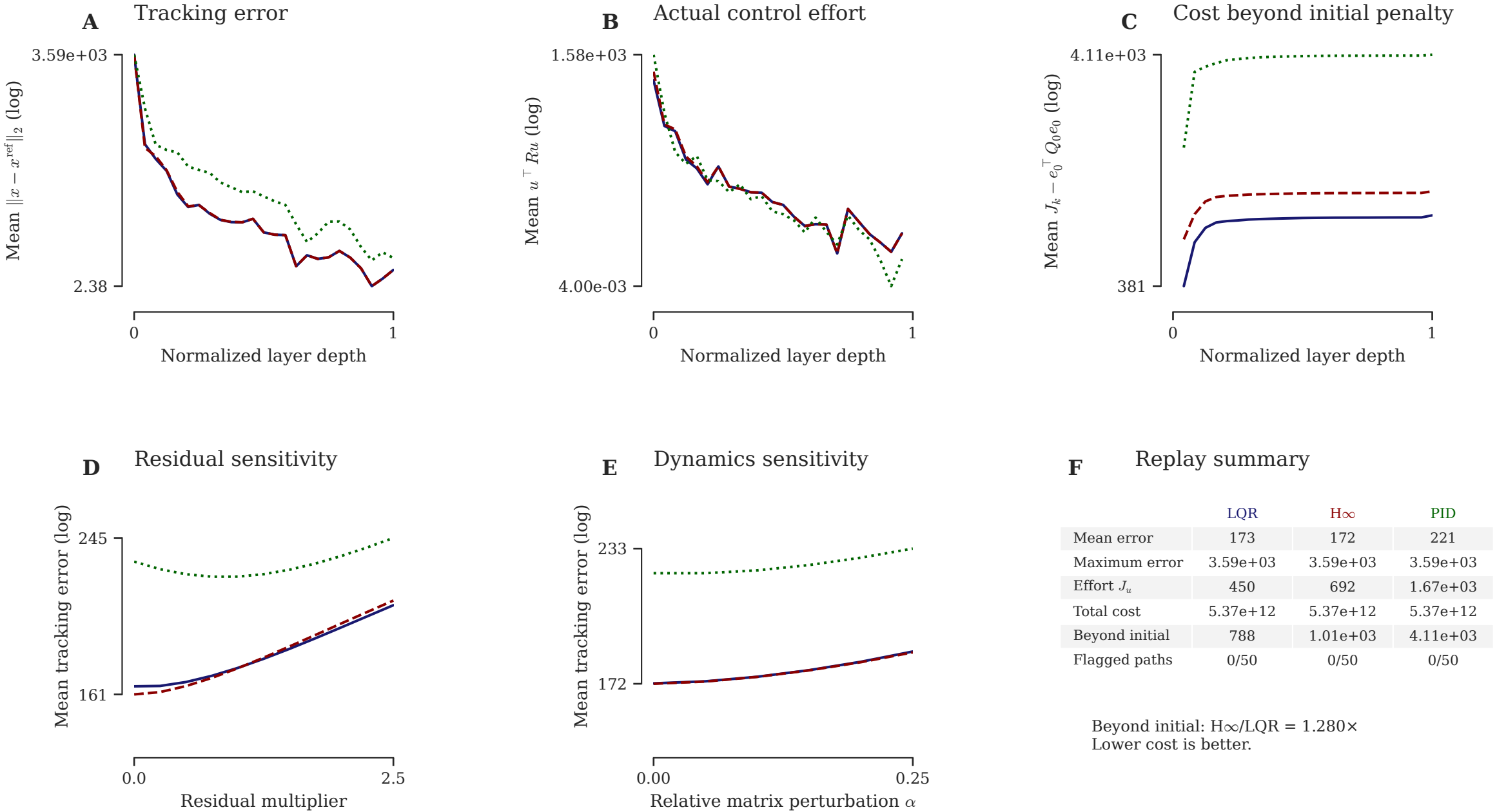


C subtracts the same fixed initial-state penalty for all controllers; full totals remain in F and the CSVs.
Costs include reference feedforward. Offline calibration replay; not recorded three-controller LM performance.

Qwen2.5 0.5B · Truthfulness calibration

Offline calibration replay · 50 matched paths · 24 layers · saved gains

LQR H ∞ PID (saved P-only tuning)

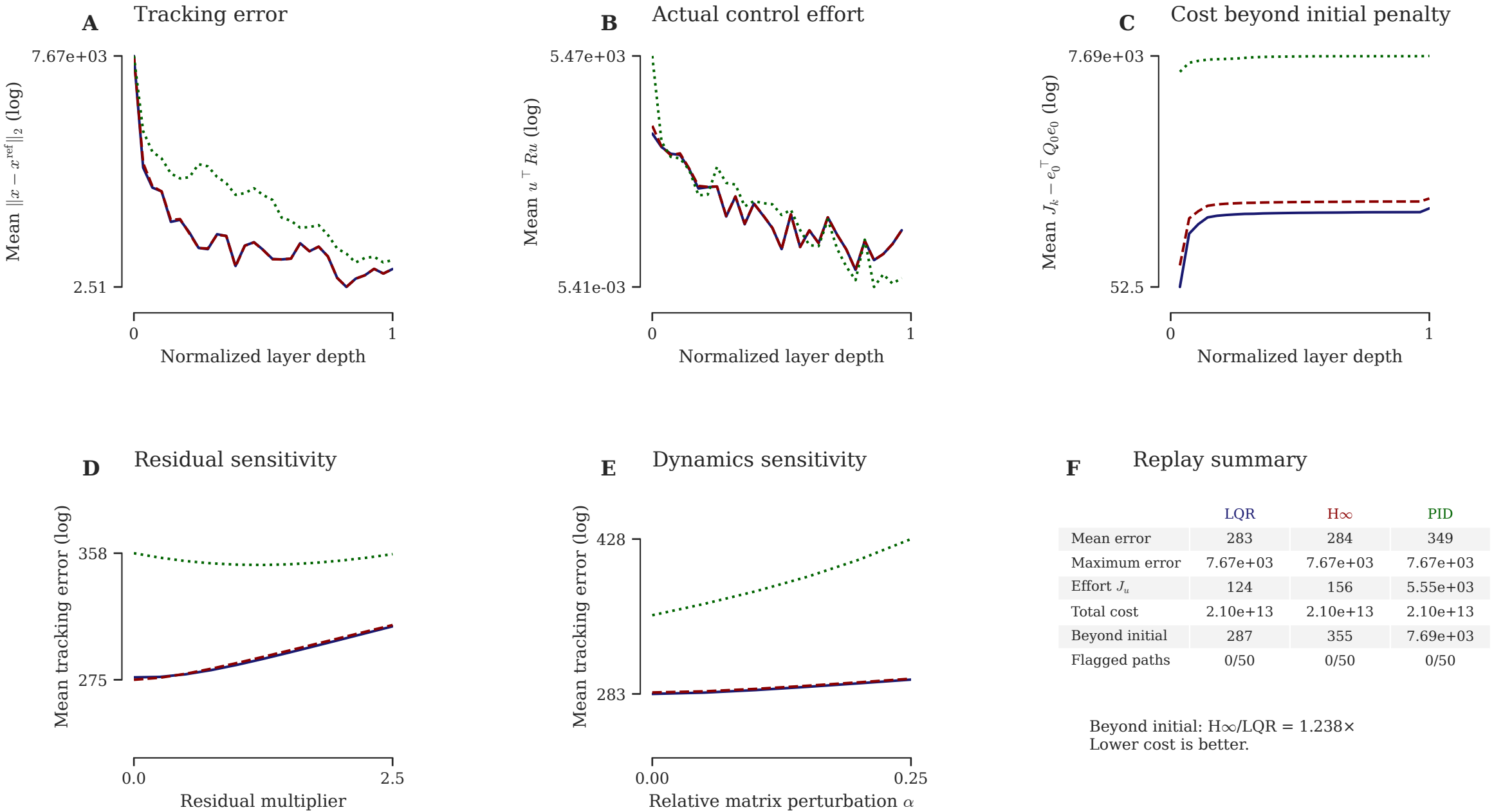


C subtracts the same fixed initial-state penalty for all controllers; full totals remain in F and the CSVs.
Costs include reference feedforward. Offline calibration replay; not recorded three-controller LM performance.

Qwen2.5 1.5B · Truthfulness calibration

Offline calibration replay · 50 matched paths · 28 layers · saved gains

LQR H ∞ PID (saved P-only tuning)

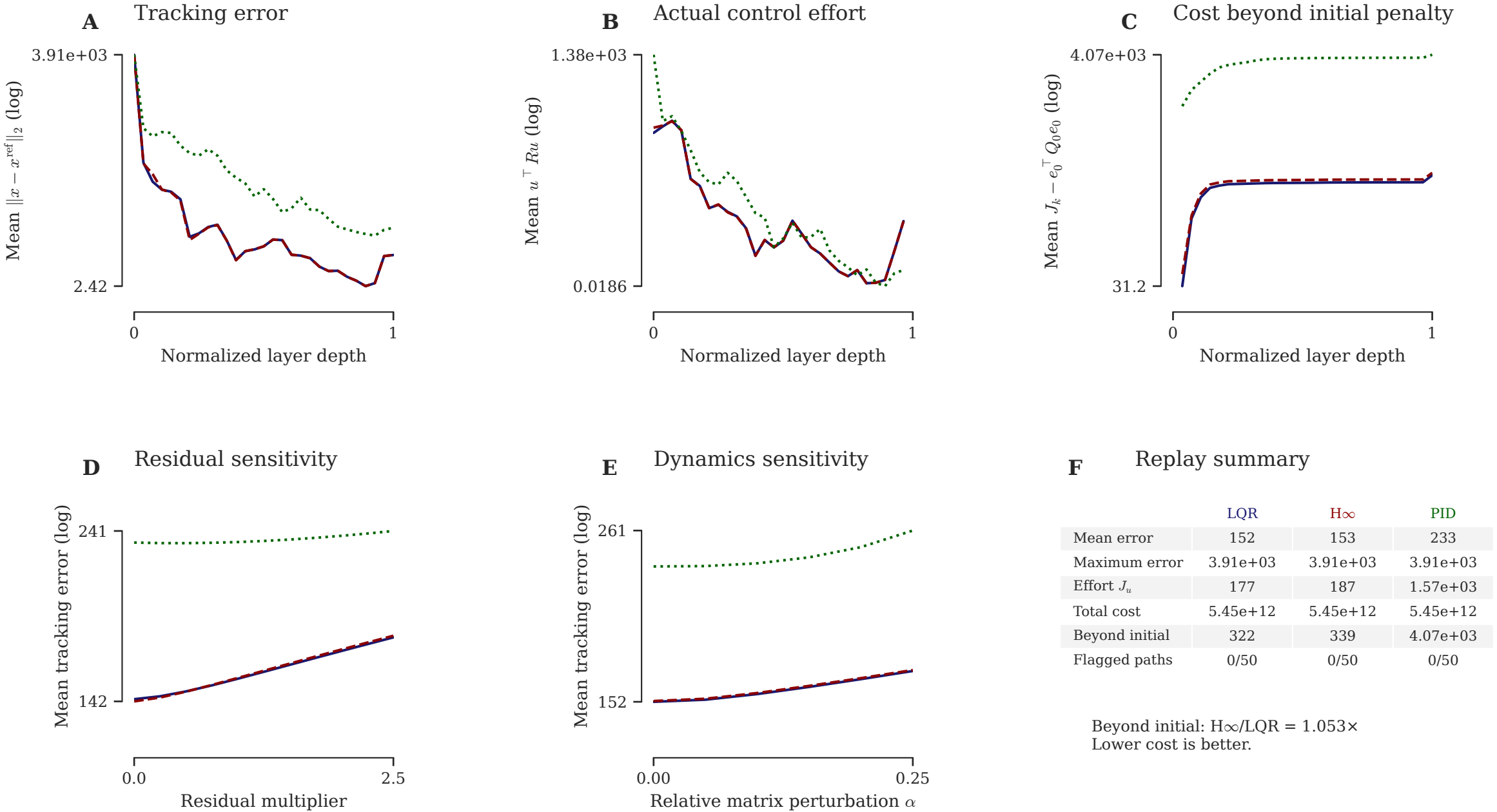


C subtracts the same fixed initial-state penalty for all controllers; full totals remain in F and the CSVs.
Costs include reference feedforward. Offline calibration replay; not recorded three-controller LM performance.

Qwen2.5 7B · Truthfulness calibration

Offline calibration replay · 50 matched paths · 28 layers · saved gains

LQR H ∞ PID (saved P-only tuning)

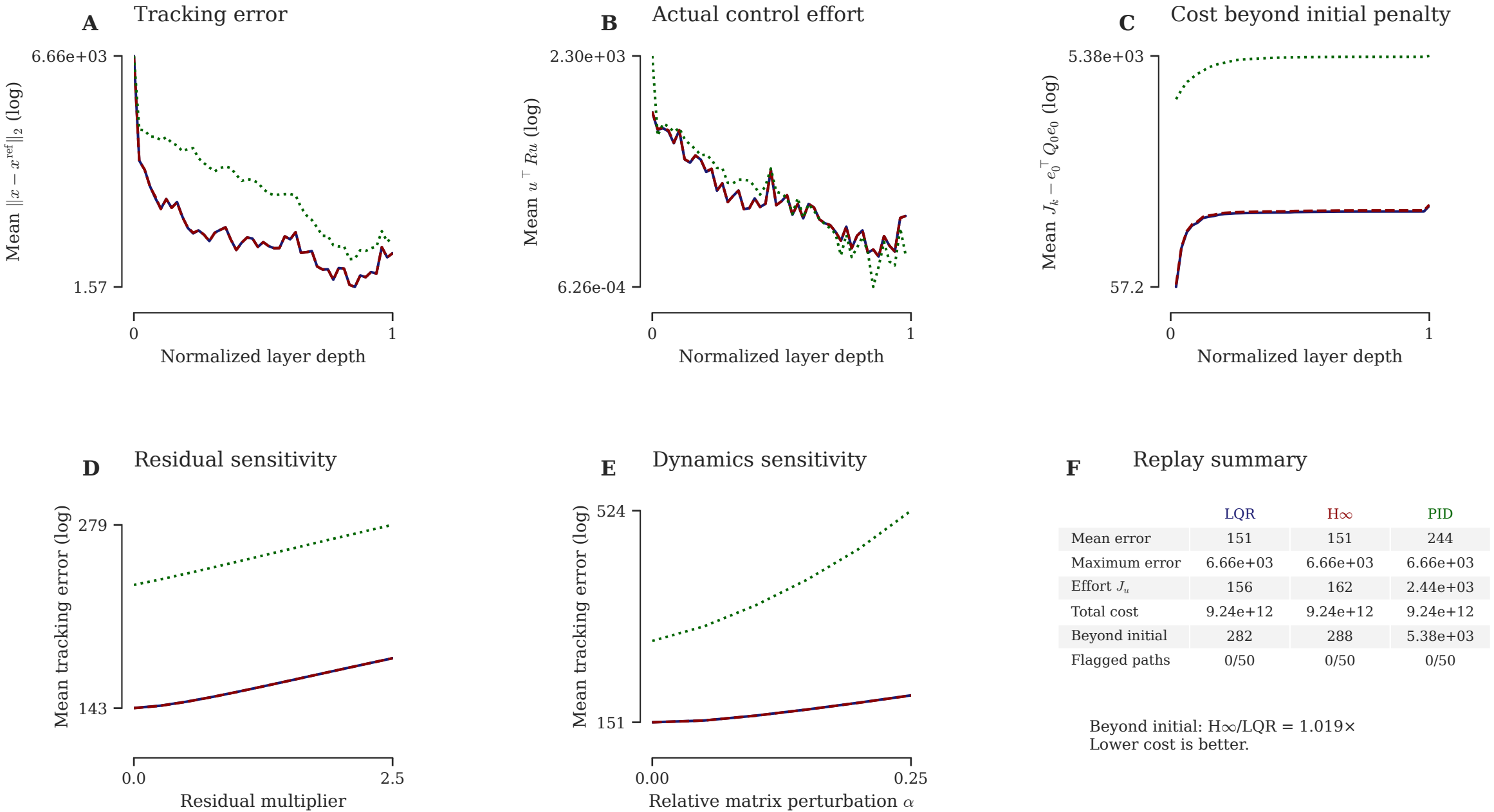


C subtracts the same fixed initial-state penalty for all controllers; full totals remain in F and the CSVs.
Costs include reference feedforward. Offline calibration replay; not recorded three-controller LM performance.

Qwen2.5 14B · Truthfulness calibration

Offline calibration replay · 50 matched paths · 48 layers · saved gains

— LQR - - - H_∞ ···· PID (saved P-only tuning)



C subtracts the same fixed initial-state penalty for all controllers; full totals remain in F and the CSVs.
Costs include reference feedforward. Offline calibration replay; not recorded three-controller LM performance.